

Answer to the question no: 02

symbol table

name	type	size	Dimension	line of Declaration	line of use	Address
x	float	bit	0	1	5 → 18	memory
v	float	bit	0	2	—	memory
m	int	10	2	3	16 → 13 →	memory
ogcount	int	bit	0	4	—	memory
char orgValues	char	bit 0	1	4	—	memory
t	float	bit	0	7	—	memory
n	int	bit	0	8	9	memory
i	int	bit	0	9	13	memory
s	float	bit	0	10	21	memory
e	int	bit	0	11	12	memory
j	int	bit	0	12	13	memory
v	int	bit	0	13	16	memory

count	int	bit	0	14	[16]	memory
ro	float	bit	0	15	[19] [21]	memory
m	float	bit	0	18	[19]	memory

Scopes

Global Symbols		
Name	Type	Data Type
x	variable	float
v	variable	float
int		
m	array	int
main	function	void
orgcount	variable	int

Name	type	Data type
i	variable	int
s	float variable	float
e	int variable	int
t	variable	float
ro	variable	int

	name	type	Data type
level 2	s	variable	float
	c	variable	int
	j	variable	int

	name	type	Date type
level 3	v	variable	int
	count	variable	int
	n	variable	float

Answer no: 01

SDT

06

record { float [2][3] a; int b } P;

P → M D

M → E { offset = 0; }

D_i → T id { top.put (id.lexeme, T.type, offset);
offset = offset + T.width }

D_i

D → E

T → record '{' { Env.push(top); top = new Env();
Stack.push(offset); offset = 0; }

D '}'

{ T.type = record(top); T.width = offset; }

top = Env.pop(); offset = Stack.pop();

T → B { T = B.type; w = B.width; }

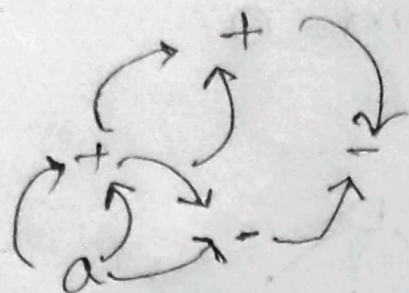
T.type = C.type; T.width = C.width; }

B → int

B → float

{ B.type = integer; B.width = 4; }

Answer no:3



$$(a+a) + (a+a) - ((a+a) - a)$$

$$t_1 = a$$

$$t_2 = a$$

$$t_3 = t_1 + t_2$$

$$t_4 = \text{unminus } t_1$$

$$t_5 = t_3 + t_4$$

$$t_6 = t_3$$

$$t_7 = t_3$$

$$t_8 = t_6 + t_7$$

$$t_9 = \text{unminus } t_5$$

$$t_{10} = t_8 + t_9$$

	OP	arg1	arg2	result
(0)	=	a		t ₁
(1)	=	a		t ₂
(2)	+	t ₁	t ₂	t ₃
(3)	unminus	t ₁		t ₄
(4)	+	t ₃	t ₄	t ₅
(5)	=	t ₃		t ₆
(6)	=	t ₃		t ₇
(7)	+	t ₆	t ₇	t ₈
(8)	unminus	t ₅		t ₉
(9)	+	t ₈	t ₉	t ₁₀

Answer: 02

offset = 0

